Dat Nguyen

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Education

University of Texas at Arlington | Arlington, TX Expected Dec 2026
Bachelor of Science in Computer Science *GPA 3.6*

Tarrant County College | Fort Worth, TX May 2024
Associate of Science *GPA 3.9*

Relevant Coursework
Data Structures and Algorithms, Databases, Fundamentals of Software Engineering, Machine Learning, Object-Oriented Programming

Experience

Student Software Developer — ACM Create Sep 2026 – Present
NoteWave — Team-built audio-to-note transcription web app *University of Texas at Arlington*

- Developed **FastAPI** endpoints for MP3 uploads, validating file signatures and rejecting unsupported files before processing
- Implemented **asynchronous background jobs** with RQ workers and job IDs so clients can poll processing status
- Added **SHA-256-based result caching** with Redis and expiration to reuse cached results across API restarts
- Configured **CORS** to support requests from the team's React/Vite frontend

Projects

Book Marketplace REST API *Java, Spring Boot, PostgreSQL, Spring Data JPA, JUnit, Mockito*

- Built a used-book marketplace **REST API** with controller, service, and repository layers, constructor-based dependency injection, and **PostgreSQL** persistence
- Modeled users, book listings, and orders with **Spring Data JPA** relationships and implemented purchase rules with appropriate 404 and 409 HTTP responses
- Prevented duplicate purchases during concurrent requests using **optimistic locking** with JPA `@Version`, returning 409 Conflict to the losing buyer
- Wrote 12 tests with **JUnit 5**, **Mockito**, and **MockMvc** covering purchase conflicts and a regression where POST requests could overwrite existing records

Dungeon Crawler Game *C#, Unity, Git*

- Built a **procedural dungeon generator** using binary space partitioning (BSP), seeded layouts, and breadth-first search (BFS) to select start and boss rooms
- Implemented **room-based encounters** with enemy counts scaled to room area, encounter-triggered spawning, and doors that reopen after enemies are defeated
- Integrated boss encounters, win/death messages, and scene transitions between the dungeon and overworld in a five-person team project

Arlington Organic Market Database *Python, SQL, HTML, MySQL*

- Designed an EER model and normalized relational schema using **SQL** for a market inventory management database
- Built a lightweight interface using **Python** and **HTML** to display and update records stored in **MySQL**

Technical Skills

Languages: Java, C, C++, C#, Python, JavaScript, HTML, CSS, SQL
Frameworks/Engines: Spring Boot, Spring Data JPA, FastAPI, Flutter, Unity, Raylib
Background Jobs/Caching: RQ, Redis
Testing: JUnit 5, Mockito, MockMvc
Databases: PostgreSQL, MySQL
Developer Tools: Git/GitHub, Maven, VS Code, Visual Studio, curl
Operating Systems: Windows, Linux, macOS

Additional

U.S. Citizen | Eligible for DoD Security Clearance